

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (AIML) (Credit System) Degree Examinations

January 2023

20AM503 - PATTERN AND VISUAL RECOGNITION

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) What is the Probability Density Function? Explain the types of variables. How to find the Probability Density Function in Statistics?
- b) Differentiate between Statistical pattern recognition with Syntactic pattern recognition.
- c) i) In the following Figure 1, 8 Patterns are considered. Find the minimum spanning tree for the 8 patterns.
ii) Consider a 4x4 square, where each of the square is a pixel to represent a digit. The squares are assigned an alphabet as shown in Figure 2. Show, how the digits 0, 1, 7, 9 and 4 can be represented using 4x4 square.

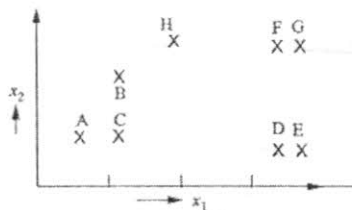


Figure 1

a	b	c	d
e	f	g	h
i	j	k	l
m	n	o	p

Figure 2

2. a) "Feature selection can also be used to speed up the process of classification, at the same time, ensuring that the classification accuracy is optimal". Justify your answer.
- b) Consider a data set with patterns having four features f1, f2, f3 and f4. The criterion function value using up to two features at a time is shown below. If J(f1) represents the criterion function using only feature f1, then J(f1) = 10; J(f2) = 20; J(f3) = 30; J(f4) = 40. This is depicted by the diagonal entries shown in below Table.

Criterion function using a sub-set of features

Feature	f1	f2	f3	f4
f1	10	30	35	55
f2	30	20	40	53
f3	35	40	30	42
f4	55	53	42	40

Show the calculation using Max – Min Approach to Feature Selection of the above features.

- c) Define pattern recognition. Discuss the applications of pattern recognition.
3. a) "Classification is a critical problem in data science and machine learning. The KNN is one of the oldest yet accurate algorithms used for pattern classification and regression models." Justify your answer.

6 L3 1 2

6 L2 2 1

8 L3 2 2

6 L2 1 5

5 L2 2 1

- b) Consider the set of two - dimensional patterns:

(1,1,1) (1,2,1) (1,3,1) (2,1,1) (2,2,1) (2,3,1)
 (2,3,5,1), (2,5,2,1) (3,5,1,1) (3,5,2,1), (3,5,3,2)
 (3,5,4,2), (4,5,1,2), (4,5,2,2), (4,5,3,2)
 (5,4,2), (5,5,2), (6,3,2), (6,4,2), (6,5,2)

where each pattern is represented by feature 1, feature 2 and the class.

- i) If a test pattern P is at (3.8, 3.1) find the class of P using nearest - neighbor algorithm.

- ii) Find the class of P using the KNN, where k is 3.

- iii) Find class of P using mKNN where K is 3.

- c) Briefly discuss the Condensed Nearest Neighbour Algorithm.

8 L3 1 5
7 L1 2 2

Unit – II

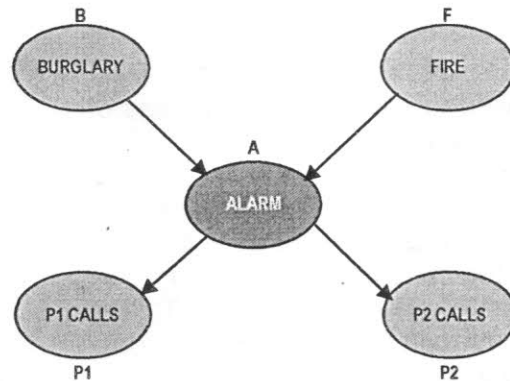
4. a) Consider the following Table. Suppose only the first ten patterns are there in the data set. Apply the Naïve Bayes Classifier to classify the given test pattern 'Cook = Sita, Mood = Bad, and Cuisine = Continental.'

Example training data set.

Cook	Mood	Cuisine	Tasty
Sita	Bad	Indian	Yes
Sita	Good	Continental	Yes
Asha	Bad	Indian	No
Asha	Good	Indian	Yes
Usha	Bad	Indian	Yes
Usha	Bad	Continental	No
Asha	Bad	Continental	No
Asha	Good	Continental	Yes
Usha	Good	Indian	Yes
Usha	Good	Continental	No

8 3 2 2

- b)



In the above Figure 4 b), we have an alarm 'A' – a node, say installed in a house of a person 'gfg', which rings upon two probabilities, i.e. burglary 'B' and fire 'F', which are – parent nodes of the alarm node. The alarm is the parent node of two probabilities P1 calls 'P1' & P2 calls 'P2' person nodes.

Upon the instance of burglary and fire, 'P1' and 'P2' call person 'gfg', respectively But, there are few drawbacks in this case, as sometimes 'P1' may forget to call the person 'gfg', even after hearing the alarm, as he has a tendency to forget things, quick. Similarly, 'P2', sometimes fails to call the person 'gfg', as he is only able to hear the alarm, from a certain distance.

Burglary 'B' -

 $P(B=T)=0.001$ ('B' is true i.e burglary has occurred) $P(B=F)=0.999$ ('B' is false i.e burglary has not occurred)

Fire 'F' -

 $P(F=T)=0.002$ ('F' is true i.e. fire has occurred) $P(F=F)=0.998$ ('F' is false i.e. fire has not occurred)

Alarm 'A' -

B	F	$P(A=T)$	$P(A=F)$
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

Person 'P1' -

A	$P(P1=T)$	$P(P1=F)$
T	0.95	0.05
F	0.05	0.95

Person 'P2' -

A	$P(P2=T)$	$P(P2=F)$
T	0.80	0.20
F	0.01	0.99

Find the probability that 'P1' is true (P1 has called 'gfg'), 'P2' is true (P2 has called 'gfg') when the alarm 'A' rang, but no burglary 'B' and fire 'F' has occurred.

6 L3 2 2

c) Write a note on Naïve - Bayes probabilistic Model.

6 L3 4 1

5. a) Write the advantages and disadvantages of Genetic algorithm.

6 L2 3 1

b) What is Genetic Algorithm? Briefly explain the outline of the basic Genetic Algorithm.

6 L2 3 1

c) What is encoding? Discuss the following with an example.

i) Permutation encoding ii) Value encoding iii) Tree encoding

8 L2 3 1

6. a) With a neat sketch, explain the stages in clustering.

6 L2 4 1

b) Apply the K-Mean clustering for following data sets (see below Table) for two clusters and tabulate all the assignments.

Sample No	X	Y
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77

10 L3 4 2

c) Briefly discuss the leader algorithm.

4 L2 4 1

Unit - III

7. a) Write an algorithm for Gradient -Based Edge Detection.

6 L2 5 1

b) Write the steps to calculate HOG Features for given image

8 L2 5 1

c) Write an algorithm for Background Subtraction.

6 L2 5 1

8. a) Briefly discuss a variety of standard techniques for computing a distance.

6 L2 5 1

b) Write an algorithm for Mean Shift Clustering.

6 L2 5 1

c) Write an algorithm for Sliding Window Detection.

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
